

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO3 –Poster Presentation	Assessment:	Assessment Tool 1– Poster Presentation
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	Mythrea Shree	Reg. No.:	192424089
Section / Batch:	BTECH AIDS	Date:	18/08/2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative,	Logical	Basic solutions	Incorrect or no	

		feasible solutions with strong justification	solutions with adequate justification	with weak justification	solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Descriptive Statistics, Sampling & Data Analysis

1. Data Summarization

Given salary dataset (₹'000): 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120.

(a) Mean and Median

$$\text{Mean} = \Sigma X / n$$

Sum of salaries:

$$32 + 35 + 36 + 38 + 40 + 42 + 45 + 48 + 50 + 95 + 120 = 581$$

$$\text{Mean} = 581 / 11 = 52.818... \approx 52.82 \text{ thousand}$$

Therefore, mean salary $\approx$ ₹52,820.

For 11 ordered observations, the median is at position $(n + 1)/2 = 6$.

$$\text{Median} = 6\text{th observation} = 42 \text{ thousand} = ₹42,000$$

Measure	Value
Mean	₹52.82 thousand
Median	₹42 thousand

(b) Why the mean is poor here

The values ₹95,000 and ₹1,20,000 are much larger than the other salaries. The mean uses every observation and is therefore pulled upward by these extreme values. The median is resistant to such outliers and better represents the typical employee in this dataset.

$$\text{Mean} = 52.82 \text{ thousand, whereas Median} = 42 \text{ thousand}$$

Because most salaries lie between ₹32,000 and ₹50,000, the median gives a more realistic measure of central tendency.

(c) SD or IQR?

I would trust the IQR more. Standard deviation squares deviations from the mean, so extreme salaries can have a disproportionately large effect on it. IQR measures the spread of the middle 50% and is much more resistant to outliers.

$$\text{IQR} = Q3 - Q1$$

Conclusion: IQR is the more robust dispersion measure for this outlier-heavy salary dataset.

2. Outlier Treatment — $1.5 \times \text{IQR}$ Rule

Step 1: Find Q1 and Q3

Median = 42. Excluding the median, the lower half is 32, 35, 36, 38, 40, so $Q1 = 36$. The upper half is 45, 48, 50, 95, 120, so $Q3 = 50$.

$$Q1 = 36; \quad Q3 = 50$$

Step 2: IQR

$$\text{IQR} = Q3 - Q1 = 50 - 36 = 14$$

Step 3: Fences

$$\text{Lower fence} = Q1 - 1.5(\text{IQR}) = 36 - 1.5(14) = 36 - 21 = 15$$

$$\text{Upper fence} = Q3 + 1.5(\text{IQR}) = 50 + 1.5(14) = 50 + 21 = 71$$

Quantity	Value (₹'000)
Q1	36
Q3	50
IQR	14
Lower fence	15
Upper fence	71

Step 4: Identify outliers

Any observation below 15 or above 71 is an outlier. Thus 95 and 120 qualify.

$$95 > 71 \rightarrow \text{outlier}$$

$$120 > 71 \rightarrow \text{outlier}$$

Winsorization

If the outliers are capped at the upper fence rather than removed, replace 95 and 120 by 71.

$$\text{New data} = 32, 35, 36, 38, 40, 42, 45, 48, 50, 71, 71$$

$$\text{New total} = 581 - (95 - 71) - (120 - 71) = 581 - 24 - 49 = 508$$

$$\text{Winsorized mean} = 508 / 11 = 46.18 \text{ thousand}$$

The mean falls from 52.82 thousand to about 46.18 thousand. Qualitatively, winsorization pulls the mean downward because the extreme high values have been capped, while retaining the observations.

3. Measuring Asymmetry

Both datasets have mean = 50 and SD = 10. The difference is skewness.

Dataset A: Pearson skewness = +1.2

Positive skewness means a long right-hand tail. Most observations are concentrated toward the lower/central side, with a smaller number of unusually high observations stretching to the right.

Typical positive-skew order: Mean > Median > Mode

Sketch description: a tall cluster on the left/centre followed by a long thin tail extending toward large values.

Dataset B: Pearson skewness = -0.8

Negative skewness means a long left-hand tail. Most observations are concentrated toward the higher-value side, with a smaller number of low observations stretching toward the left.

Typical negative-skew order: Mean < Median < Mode

Sketch description: a tall cluster on the right/centre followed by a long tail extending toward small values.

Which is more salary-like?

Dataset A. Salary distributions commonly have a long right tail because most employees earn moderate salaries while a smaller group of senior employees earn very high salaries.

More salary-like distribution → Dataset A, because skewness = +1.2

4. Coefficient of Skewness Calculation

Given: Mean = 62, Median = 58, Mode = 52, SD = 12.

(a) Karl Pearson's coefficient — mode method

$$Sk = (\text{Mean} - \text{Mode}) / SD$$

$$Sk = (62 - 52) / 12 = 10/12 = 0.8333 \approx +0.83$$

The positive value indicates positive skewness.

(b) Karl Pearson's coefficient — median method

$$Sk = 3(\text{Mean} - \text{Median}) / SD$$

$$Sk = 3(62 - 58) / 12 = 12/12 = +1.00$$

Comparison

Method	Skewness
Mode method	+0.83
Median method	+1.00

The results differ slightly because the formulas use different measures of central tendency. In a skewed distribution, mean, median, and mode are not equal. Here, $62 > 58 > 52$, which consistently indicates positive skewness.

5. Variance vs Standard Deviation Interpretation

Line A: Mean = 200 units/day, Variance = 64. Line B: Mean = 150 units/day, Variance = 36.

Step 1: Standard deviation

$$SD = \sqrt{\text{Variance}}$$

$$\text{Line A: } SD = \sqrt{64} = 8$$

$$\text{Line B: } SD = \sqrt{36} = 6$$

Step 2: Coefficient of Variation

$$CV = (SD / \text{Mean}) \times 100$$

$$\text{Line A: } CV = (8/200) \times 100 = 4\%$$

$$\text{Line B: } CV = (6/150) \times 100 = 4\%$$

Although Line A has the larger raw variance (64 versus 36), both lines have exactly the same relative variability.

$$CV_A = CV_B = 4\%$$

Conclusion: neither line has greater relative variability; both are equally variable relative to their respective means.

6. Sampling Distribution Reasoning

Population mean $\mu = 500$, population SD $\sigma = 80$, sample size $n = 64$.

(a) Standard error

$$SE_{\bar{X}} = \sigma / \sqrt{n} = 80 / \sqrt{64} = 80/8 = 10$$

$$\text{Answer: } SE = 10$$

(b) $P(\bar{X} > 520)$

$$Z = (\bar{X} - \mu) / SE = (520 - 500) / 10 = 2$$

From the standard normal distribution, $P(Z < 2) = 0.9772$.

$$P(Z > 2) = 1 - 0.9772 = 0.0228 = 2.28\%$$

Therefore, there is a 2.28% probability that the sample mean exceeds 520, under the stated normal approximation.

(c) If $n = 256$

$$SE = 80 / \sqrt{256} = 80/16 = 5$$

The SE falls from 10 to 5, i.e. it is halved. The reason is the square-root relationship:

$$SE \propto 1/\sqrt{n}$$

Increasing sample size makes the sample mean more precise and reduces sampling error.

7. Covariance Calculation and Sign Interpretation

X (₹'000): 10, 20, 30, 40, 50. Y (₹ lakh): 15, 18, 25, 30, 42.

$$\bar{X} = (10+20+30+40+50)/5 = 30$$

$$\bar{Y} = (15+18+25+30+42)/5 = 26$$

X	Y	X- $\bar{X}$	Y- $\bar{Y}$	Product
10	15	-20	-11	220
20	18	-10	-8	80
30	25	0	-1	0
40	30	10	4	40
50	42	20	16	320

$$\Sigma(X-\bar{X})(Y-\bar{Y}) = 220 + 80 + 0 + 40 + 320 = 660$$

$$\text{Sample Cov}(X,Y) = 660/(n-1) = 660/4 = 165$$

Answer: sample covariance = +165. (If a population covariance convention is used, divide by n and obtain 132.)

The positive sign is intuitive: higher advertising spend is associated with higher sales.

However, covariance magnitude alone cannot tell us how strong the relationship is because covariance depends on the measurement units. Correlation standardizes the relationship to a scale from -1 to +1 and therefore communicates strength and direction more directly.

8. Correlation vs Causation Trap

Given $r = 0.85$ between ice-cream sales and drowning incidents.

Statistical flaw

A correlation of +0.85 indicates a strong positive association, not proof that ice cream causes drowning.

Correlation $\neq$ Causation

Likely confounding variable

Temperature or hot weather is a classic confounder. Hot weather increases ice-cream consumption and also encourages swimming and other water activities, which can increase drowning risk.

Hot weather → more ice-cream sales

Hot weather → more swimming → more drowning incidents

What evidence is needed for causation?

- Control or adjust for confounding variables such as temperature, season, holidays, and swimming activity.
- Establish temporal order: the proposed cause must occur before the effect.
- Use strong experimental evidence where ethical and feasible, or strong observational designs when experiments are impossible.
- Replicate the finding across locations, years, and populations.

Thus $r = 0.85$ is evidence of association, not evidence that ice cream causes drowning.

9. Effect of Outliers on Correlation

Initially $r = 0.42$. After removing one extreme outlier, $r = 0.89$.

This large change shows that Pearson's correlation can be highly sensitive to influential observations. Pearson's r uses the actual numerical values and deviations from the means, so a single extreme point can substantially change the calculated relationship.

0.42 → 0.89 after removing one point

The outlier was masking a much stronger positive linear association among the remaining observations.

Would Spearman be more robust?

Generally, yes. Spearman's rank correlation uses ranks rather than raw magnitudes, so an extreme numerical value has less influence. It is therefore more robust to outliers than Pearson's r , although it is not completely immune to influential observations.

Pearson	Spearman
Uses actual values	Uses ranks
Measures linear association	Measures monotonic association
More sensitive to extreme magnitudes	Generally more robust to extreme magnitudes

10. Data Preparation — Missing Values

There are 200 rows and 15% missing Income values. Missingness is more common among unemployed respondents.

(a) Classify the mechanism

If employment status is an observed variable in the dataset, this is best classified as MAR (Missing At Random), because missingness is related to an observed characteristic.

Missing Income related to observed Employment Status → MAR

Do not confuse 'not random' in ordinary language with MNAR. MNAR would mean missingness depends on the missing income itself or another unobserved factor.

(b) Why mean imputation can be biased

Overall mean imputation assumes the missing observations resemble the overall population. That is questionable because missing income is disproportionately associated with unemployment. If unemployed respondents have systematically different incomes, assigning them the overall mean can artificially raise or otherwise distort their incomes and can bias relationships involving income.

(c) Better strategy

A better approach is group-wise imputation using employment status, or preferably multiple imputation using employment status and other predictive variables such as age, education, occupation, and experience. Multiple imputation also reflects uncertainty about the missing values instead of pretending every missing value is known exactly.

11. Standardization / Normalization

Age ranges from 18–75, while Income ranges from ₹20,000–₹20,00,000. The data are intended for a distance-based algorithm.

Why scaling matters

Distance-based algorithms such as KNN and k-means compare observations using distances. If Income is left in rupees while Age is measured in years, the numerical scale of Income is thousands or millions of times larger than Age.

For example, a difference of 10 years in Age contributes only 10 to an unscaled Euclidean distance, whereas a ₹1,00,000 income difference contributes 100,000. The algorithm would therefore behave as if income is overwhelmingly more important, even if the analyst did not intend that.

$$\text{Euclidean distance} = \sqrt{[\sum (x_i - y_i)^2]}$$

Standardization puts variables on a comparable scale, commonly with mean 0 and SD 1. Min-max normalization instead maps values to a bounded interval such as [0,1].

Z-score for Age = 45

$$z = (X - \text{Mean}) / \text{SD}$$

$$z = (45 - 40) / 12 = 5/12 = 0.4167 \approx 0.42$$

So Age = 45 is about 0.42 standard deviations above the mean.

12. Distribution Shape from Summary Statistics Alone

Given Mean = Median = Mode = 45, skewness ≈ 0 , and kurtosis = 5.8.

Equal mean, median, and mode together with near-zero skewness suggest a distribution that is approximately symmetric around 45. However, skewness mainly describes asymmetry; it does not tell us how heavy the tails or how peaked the distribution are.

Kurtosis = 5.8 is high relative to the normal distribution's kurtosis of 3 under the ordinary (non-excess) definition. This indicates a leptokurtic distribution: relatively heavy tails and a sharper concentration/peak than a normal distribution.

Shape summary: approximately symmetric + leptokurtic (heavy-tailed)

Thus, the central tendency and skewness alone could make the distribution look 'normal-like', but the high kurtosis warns that extreme observations may occur more often than in a normal distribution.

13. Grouped Data — Mean and Variance by Step-Deviation

Class interval	Frequency f	Midpoint x
0–10	5	5
10–20	8	15
20–30	15	25
30–40	12	35
40–50	10	45

Choose assumed mean $A = 25$ and class width $h = 10$.

$$u = (x - A)/h$$

x	f	u	fu	fu ²
5	5	-2	-10	20
15	8	-1	-8	8
25	15	0	0	0
35	12	1	12	12
45	10	2	20	40

$$N = \Sigma f = 5+8+15+12+10 = 50$$

$$\Sigma fu = -10-8+0+12+20 = 14$$

$$\Sigma fu^2 = 20+8+0+12+40 = 80$$

Mean

$$\text{Mean} = A + h(\Sigma fu/N)$$

$$\text{Mean} = 25 + 10(14/50) = 25 + 2.8 = 27.8$$

$$\text{Mean} = 27.8$$

Variance

$$\text{Variance} = h^2[(\Sigma fu^2/N) - (\Sigma fu/N)^2]$$

$$= 10^2[(80/50) - (14/50)^2]$$

$$= 100[1.6 - 0.0784] = 100(1.5216) = 152.16$$

$$\text{Variance} = 152.16$$

Therefore, the grouped-data mean is 27.8 and the population-style grouped variance is 152.16. If a sample variance convention using $N-1$ is required, the numerical variance would be slightly larger.

14. Comparing Two Distributions via CV

Portfolio A: Mean return = 12%, SD = 3%. Portfolio B: Mean return = 18%, SD = 5%.

$$CV = SD / \text{Mean} \times 100$$

Portfolio A

$$CV_A = (3/12) \times 100 = 25\%$$

Portfolio B

$$CV_B = (5/18) \times 100 = 27.78\%$$

Portfolio	Mean	SD	CV
A	12%	3%	25%
B	18%	5%	27.78%

Portfolio B is riskier in absolute terms because its SD is higher (5% versus 3%). However, it also has a higher expected return. CV measures risk relative to the amount of return: it is the amount of variability per unit of mean return.

Because B has $CV \approx 27.78\%$, compared with A's 25%, B has slightly higher relative risk per unit of return. Therefore, the investor's first part is correct, but the 'not necessarily worse per unit of return' claim should be qualified: B is actually slightly worse on this specific CV measure.

15. Outlier's Effect on Variance

Original dataset: 10 values, variance = 25. One value 40 is mistakenly recorded as 400. The original value 40 is close to the mean 42.

(a) Effect on mean

The value increases by:

$$400 - 40 = 360$$

$$\text{Mean increase} = 360/10 = 36$$

So the mean moves upward substantially, from about 42 toward about 78, if all other values are unchanged.

(b) Effect on variance

Variance is affected even more severely because it is based on squared deviations from the mean.

The original value 40 was only about 2 units below the mean, so its squared deviation was tiny:

$$(40 - 42)^2 = 4$$

The erroneous value 400 is hundreds of units away from the original centre, so its squared deviation is enormous. Although the mean itself also shifts upward and partially reduces some deviations, the squared-distance term causes a dramatic inflation of the variance.

$$\text{Variance is based on } \Sigma(X - \bar{X})^2$$

Therefore, the variance will increase by a very large factor and the standard deviation will also rise sharply. This illustrates why variance is especially sensitive to extreme errors and outliers.

16. Continuous Distribution — Area Interpretation

$X \sim \text{Normal}(\mu = 100, \sigma = 15)$.

(a) Proportion between 85 and 130

$$\text{For } X = 85: z = (85 - 100)/15 = -1$$

$$\text{For } X = 130: z = (130 - 100)/15 = 2$$

So we need $P(-1 < Z < 2)$. From the standard normal table:

$$P(Z < 2) = 0.9772$$

$$P(Z < -1) = 0.1587$$

$$P(-1 < Z < 2) = 0.9772 - 0.1587 = 0.8185$$

$$\text{Proportion} = 0.8185 = 81.85\%$$

(b) $X = 145$

$$z = (145 - 100) / 15 = 45 / 15 = 3$$

The value is exactly 3 standard deviations above the mean.

$$z = +3$$

Under the empirical rule, about 99.7% of normal observations lie within ± 3 SD. A value at $+3$ SD is therefore unusual and is reasonable to flag for investigation as a potential outlier, although the empirical rule itself does not prove that it is an error.

17. Sampling Bias vs Sampling Error

The survey samples only online panel respondents. Sample mean = ₹18,000, while the known population mean = ₹14,500.

(a) **Bias or error?**

This is better explained by sampling bias, specifically coverage/selection bias. People who are part of an online panel may systematically differ from the full population in income, spending, internet access, age, or other characteristics.

$$\text{Observed difference} = ₹18,000 - ₹14,500 = ₹3,500$$

Sampling error is random variation that naturally occurs because a sample rather than the entire population is observed. Here, the sampling method itself excludes or underrepresents parts of the population, producing systematic error.

(b) **Would a larger sample fix it?**

No. Increasing the sample size within the same biased online-panel frame can make the estimate more precise around the wrong target. A huge biased sample is still biased.

To address the problem, the sampling frame must better represent the population, for example through probability sampling or appropriate weighting/post-stratification.

18. Covariance Sign vs Correlation Strength

Pair 1 has $\text{Cov} = 850$ and Pair 2 has $\text{Cov} = 12$. Both are positive.

The positive signs tell us both pairs have positive co-movement. But 850 cannot automatically be called stronger than 12.

Covariance depends on the units and scales of the variables. A pair measured in large units can naturally produce a much larger covariance than a pair measured in small units even if its relationship is weaker.

$$\text{Correlation } r = \text{Cov}(X, Y) / (\text{SD}_X \text{SD}_Y)$$

The missing information is the standard deviations of X and Y (or, more directly, the correlation coefficient itself). Correlation removes the scale dependence and ranges from -1 to +1.

Need: SD_X, SD_Y, or standardized correlation r

19. Estimation — 95% Confidence Interval

Given $n = 100$, sample mean = 250, sample SD = 40.

Step 1: Standard error

$$SE = s/\sqrt{n} = 40/\sqrt{100} = 40/10 = 4$$

Step 2: 95% critical value

For a large sample, use $z^* = 1.96$ for a 95% confidence interval.

$$\text{Margin of error} = 1.96 \times 4 = 7.84$$

Step 3: Confidence interval

$$CI = \text{Mean} \pm \text{Margin of Error}$$

$$CI = 250 \pm 7.84$$

$$\text{Lower limit} = 242.16$$

$$\text{Upper limit} = 257.84$$

$$95\% \text{ CI} = [242.16, 257.84]$$

Interpretation: using this sampling procedure, we construct intervals that capture the true population mean about 95% of the time in repeated samples. For this sample, the interval is from 242.16 to 257.84.

What it does NOT mean: it does not mean there is a 95% probability that the fixed population mean is inside this particular interval. The population mean is fixed; the interval is the random object produced by sampling.

It also does not say that 95% of individual observations lie between 242.16 and 257.84. A confidence interval is about the population mean, not individual data values.

20. Integrated Case Study

Daily sales (₹'000) for 15 days:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

(a) Identify the outlier using IQR and z-score

Sort the observations:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500

With 15 observations, the median is the 8th value = 124. Excluding the median, the lower half is 117–123 and the upper half is 125–500.

$$Q1 = 120$$

$$Q3 = 128$$

$$IQR = 128 - 120 = 8$$

$$\text{Lower fence} = 120 - 1.5(8) = 120 - 12 = 108$$

$$\text{Upper fence} = 128 + 1.5(8) = 128 + 12 = 140$$

$$500 > 140 \rightarrow \text{outlier}$$

Thus, the IQR method clearly identifies 500 as an outlier.

Z-score method

$$\text{Mean} = (\text{sum of all 15 values})/15 = 2227/15 = 148.47$$

Using the population-style SD for the descriptive z-score:

$$SD \approx 94.03$$

$$z \text{ for } 500 = (500 - 148.47)/94.03 \approx 3.74$$

Since $|z| \approx 3.74 > 3$, 500 is a strong potential outlier by the common $|z| > 3$ rule. Using sample SD instead gives $z \approx 3.61$, leading to the same conclusion.

(b) Mean and median with and without the outlier

With the outlier:

$$\text{Mean} = 2227/15 = 148.47 \text{ thousand}$$

$$\text{Median} = 124 \text{ thousand}$$

Without 500:

$$\text{New total} = 2227 - 500 = 1727$$

$$\text{New } n = 14$$

$$\text{Mean} = 1727/14 = 123.36 \text{ thousand}$$

$$\text{Median} = (7\text{th} + 8\text{th values})/2 = (123 + 124)/2 = 123.50 \text{ thousand}$$

Statistic	With 500	Without 500
Mean (₹'000)	148.47	123.36
Median (₹'000)	124	123.50

The mean changes by about 25.11 thousand, while the median changes by only 0.50 thousand. This demonstrates that the mean is highly sensitive to extreme values, whereas the median is robust.

(c) Skewness with the outlier included

Using Karl Pearson's median-based coefficient:

$$S_k = 3(\text{Mean} - \text{Median}) / SD$$

$$S_k = 3(148.47 - 124) / 94.03$$

$$S_k \approx 73.41 / 94.03 \approx +0.78$$

The positive skewness indicates a right-skewed distribution. The extreme value 500 creates a long right tail.

Note: a moment-based skewness statistic can produce a different numerical value because different skewness definitions exist. The important conclusion here is the strong right-tail asymmetry.

(d) Recommendation: remove, cap, or investigate?

The best first action is INVESTIGATE, not automatically remove or cap the value.

- IQR method: 500 is far above the upper fence of 140.
- Z-score method: $z \approx +3.74$, well beyond the common ± 3 screening threshold.
- Mean is pulled from 123.36 thousand without the value to 148.47 thousand with it.
- Median barely moves from 123.50 to 124 thousand, showing that the typical day is around ₹123–124 thousand.

These results strongly suggest that 500 is influential and unusual. However, unusual does not necessarily mean wrong. It could represent a genuine festival sale, bulk order, promotion, seasonal event, or data-entry error.

Therefore, investigate the source record first. If 500 is confirmed as a genuine event, retain it but report it transparently and consider robust summaries such as the median. If it is confirmed as a recording error, correct it or remove it according to the data-quality policy. Capping should be used only when there is a defensible winsorization policy, not simply because the value looks inconvenient.

Recommended decision: INVESTIGATE → validate → then retain/correct/remove according to evidence

Quick Formula Sheet

Topic	Key Formula
Mean	$\bar{x} = \Sigma x / n$
Median (odd n)	Position = $(n+1)/2$
IQR	$IQR = Q3 - Q1$
Outlier fences	$Q1 - 1.5IQR ; Q3 + 1.5IQR$
Standard deviation	$SD = \sqrt{\text{Variance}}$
Coefficient of Variation	$CV = (SD/\text{Mean}) \times 100$
Standard error	$SE = \sigma/\sqrt{n}$
Z-score	$z = (X - \mu)/\sigma$
Sample covariance	$\text{Cov}(X,Y) = \Sigma[(X-\bar{x})(Y-\bar{y})]/(n-1)$
Correlation	$r = \text{Cov}(X,Y)/(SD_X SD_Y)$
Pearson skewness — mode	$Sk = (\text{Mean} - \text{Mode})/SD$
Pearson skewness — median	$Sk = 3(\text{Mean} - \text{Median})/SD$
Z-score standardization	$z = (X - \text{Mean})/SD$
Grouped mean, step-deviation	$\bar{x} = A + h(\Sigma fu/N)$
Grouped variance	$\sigma^2 = h^2[(\Sigma fu^2/N) - (\Sigma fu/N)^2]$
95% CI (large sample)	$\bar{x} \pm 1.96(SE)$

Exam tip: Always state the interpretation after the numerical answer. In statistics, the calculation tells you what happened; the interpretation tells the examiner that you understand why it matters.